

Full Answer Key – Activities 7–12

Activity 7: Debug the Robot

BASIC — *Find the Bug*

- **Program 1 (Tea):** SEQUENCE BUG. "Drink the tea" is out of order — should be LAST.
- **Program 2 (Walking Dog):** MISSING-STEP BUG. No step to put leash on dog. Add "Put leash on the dog" between steps 1 and 2.
- **Program 3 (Alarm):** WRONG-VALUE BUG. 7:00 PM should be 7:00 AM.

CHALLENGE — *Two-Bug Hunt*

- **Part A (Juice):** Bug 1 — Step 3 is a duplicate. Delete it. Bug 2 — "Wash the glass" should be AFTER the guest finishes drinking.
- **Part B (Watering loop):** Loop runs 4 times but there are 5 plants. **Fix:** change 4 to 5. Bonus insight if kid also notices the loop doesn't move to the next plant.

BONUS Grading Note: Open-ended "Be the Bug." Accept any 5-step instruction set with a deliberate, identifiable bug.

Activity 8: Bubble Sort

BASIC — *Sort [5, 2, 8, 3]*

- **Pass 1:** [2, 5, 3, 8]
- **Pass 2:** [2, 3, 5, 8]
- **Pass 3:** [2, 3, 5, 8] *(verification, no swaps)*
- **Passes needed:** 2

CHALLENGE

- **Part A — Heights [140, 125, 150, 130, 145]:** Pass 1: [125, 140, 130, 145, 150]. Pass 2: [125, 130, 140, 145, 150]. Pass 3: verification.
- **Part B:** The INNER loop runs more times. For every ONE pass of the outer loop, the inner loop runs through every pair. Bubble sort's power (and cost) comes from nested loops.

BONUS Grading Note: Open-ended (real-world sorting).

Activity 9: Selection Sort

BASIC — *Sort [6, 3, 9, 1]*

- **After Selection 1:** [1, 3, 9, 6]
- **After Selection 2:** [1, 3, 9, 6]
- **After Selection 3:** [1, 3, 6, 9]
- **First item locked in:** 1

CHALLENGE — *Sort [8, 3, 5, 1, 4] Both Ways*

- **Bubble Sort:** Pass 1: [3, 5, 1, 4, 8]. Pass 2: [3, 1, 4, 5, 8]. Pass 3: [1, 3, 4, 5, 8].
- **Selection Sort:** After Sel 1: [1, 3, 5, 8, 4]. After Sel 2: [1, 3, 5, 8, 4]. After Sel 3: [1, 3, 4, 8, 5]. After Sel 4: [1, 3, 4, 5, 8].
- **Same final answer:** YES.
- **Part B:** Open-ended discussion. Both are valid. Selection sort has fewer swaps; bubble sort can early-exit if list is nearly sorted.

BONUS Grading Note: Open-ended (algorithm race).

Activity 10: Pattern Recognition

BASIC — *Find the Next One*

- **Pattern 1:** Triangle-circle repeating. Next: **circle**.
- **Pattern 2:** Add 5. Next: **25**.
- **Pattern 3:** A-B-C repeating. Next: **B**.
- **Pattern 4:** Double. Next: **32**.

CHALLENGE

- **A1:** Fibonacci (add previous two). Next: **13**. Type: Rule-based.
- **A2:** Differences decrease by 1 (−10, −9, −8, −7, then −6). Next: **60**. Type: Rule-based.
- **A3:** Days of the week. Next: **S** (Sunday). Type: Repeating (or Rule-based — both accepted).
- **Part B:** Open-ended.

BONUS Grading Note: Open-ended pattern hunt.

Activity 11: Treasure Hunt Grid

BASIC — *Grid 1 (one valid path — others accepted)*

MOVE → TURN RIGHT → MOVE → MOVE → TURN LEFT → MOVE → MOVE → MOVE → TURN LEFT → MOVE → MOVE = **11 commands**

CHALLENGE

- **Part A — Grid 2 with loops:** Without loops: 10 commands. With loops: TURN RIGHT → REPEAT MOVE 4 TIMES → TURN LEFT → REPEAT MOVE 4 TIMES = **4 commands**.
- **Part B — Grid 3 debug:** Bug — step 4 walks Bolt into the wall at (1,1). **Fix:** TURN RIGHT before step 4. Corrected: MOVE → MOVE → MOVE → TURN RIGHT → MOVE → MOVE → TURN LEFT → MOVE = **8 commands**.

BONUS Grading Note: Open-ended (design your own grid).

Activity 12: Design Your Own Algorithm

BASIC

Open-ended. Accept any 5-step algorithm following the design process (pick problem → list steps → look for patterns → write → test).

CHALLENGE

Open-ended. Must use at least one **loop** AND one **conditional**.

BONUS Grading Note: Open-ended (teach someone). The point is that the kid actually taught someone.

End of Answer Key.